

The algorithm operates by carrying out the following steps as specified:

2. Initialize L to the empty list
3. Add the current node to the end of L and check to see if nodes appear in L two times
4. From the given node, see if there are any unmarked outgoing arcs. If yes, go to step 5, if not, go to step 6
5. Pick an unmarked outgoing arc and mark it and go to step 3.
6. We have now reached a dead end. Remove it and go back to the previous node. Make that one the current node and go to step 3. if this node is an initial node the graph doesn't contain any cycle and algorithm terminates.

Inspect the nodes from left to right, top to bottom, first running the algorithm starting at R, then successively A, B, C, S, D, T, E, F, U, V, W and G.

STEP 2

L = []

STEP 3

L = [R] node is not appear in the list twice

STEP 4

Yes there is an unmarked outgoing arc from R to A

STEP 5

Now mark this arc from R to A with CROSS (X) and again go to STEP 3

STEP 3

L = [R, A] node is not appear in the list twice

STEP 4

Yes there is an unmarked outgoing arc from A to S

STEP 5

Now mark this arc from A to S with CROSS (X) and again go to STEP 3

STEP 3

L = [R, A, S] node is not appear in the list twice

STEP 4

NO there is no unmarked outgoing arc from S (GO TO STEP 6)

STEP 6

DEAD END. Remove the last node [S] from the list and go back to the previous node [A] and repeat STEP 3. Now L = [R, A]

STEP 3

L = [R, A] node is not appear in the list twice

STEP 4

NO there is no unmarked outgoing arc from A to S. (Because previously we have marked that arc from A to S). now go to STEP 6

STEP 6

DEAD END. Remove the last node [A] from the list and go back to the previous node [R] and repeat STEP 3. Now L = [R]

STEP 3. Now L = [R, A]

STEP 3

L = [R] node is not appear in the list twice

STEP 4

NO there is no unmarked outgoing arc from R to A. (Because previously we have marked that arc from R to A). now go to STEP 6

STEP 6

DEAD END. Remove the last node [R] from the list. As this node is an initial node so graph does not contain any cycle so algorithm terminates.

Continue with the above procedure, now, we start again at B node, from B we continue to follow outgoing arcs, at which time L = [B, T, E, V, G, U, D]. Now, we make random choice. If we pick S we come to a dead end and backtrack to D. The second time we pick T and update L to be [B, T, E, V, G, U, D, T] at which point we discover the cycle and stop the algorithm because T appears in the list twice.

.....



